

clock cycles may be required by one CPU compared to another to do the same amount of work if they belong to different families. If they belong to different micro-architectures, a 3.4 GHz model may not necessarily be faster than a 3.1 GHz model, as a famous ArsTechnica article explained some time ago (<http://goo.gl/Xuf1X>)

2. **Number of cores:** Salesmen might use the word ‘multi-tasking’ more to their benefit than yours. If you only browse the net and edit documents, even two cores will suffice. For multimedia work, you might need more. The RAM plays an important role in multi-tasking and we’ll talk about it soon. Also, as we already said, most Core i3’s and i5’s are equipped with quad-core processors (dual cores with HT) so you can pick either. Core i7’s are overkill for office work but a necessity for multimedia-heavy applications. On the AMD side, almost all Phenom II processors are either triple-cores (quad cores with one core locked out) or quad-cores and they’re cast in the same mould as lower-end Core i7’s and Core i5’s when it comes to performance.
3. **Intel Turbo Boost / AMD Turbo Core:** Though the terms may differ, both technologies function in more or less the same way. Turbo Boost and Turbo Core increase the clock speed of the processor when workload increases and decrease it with low load. The key benefit of this technology is that it saves battery life. If you can, go for a laptop with a Turbo Boost enabled processor.
4. **Max TDP:** Max TDP or Thermal Design Power is roughly the amount of power the processor consumes. If your priority is battery life and not performance, go for the processors with lower TDP.
5. **Max Memory Support:** This is the amount of memory (RAM) that the processor is designed to address. It determines how much RAM you can add to the laptop. In most cases you won’t be installing more RAM than the processor limits but it’s always better to be sure. E.g. Core i3-2310M can address 16 GB of memory while Core i5-2557M supports only 8 GB!
6. **Graphics Processing Unit:** Graphics processing is one of the system’s most intensive jobs for a couple of reasons:
 - a. Graphics processing is computationally intensive by nature. The computation required for calculating the position and colour of each pixel among a million others is high. This is the reason graphics are not processed on the CPU, but on a separate unit.
 - b. Graphics processing cannot be stopped. As long as you’re using